

CI Observability, Manual Heal, and Auto-Heal Field Requirements

Purpose

This document defines the minimum operational data that each Configuration Item (CI) type should expose to support:

- reliable observability
- fast manual remediation
- safe and effective auto-heal workflows

The goal is to standardize the payload expected from monitoring and event-management tools so downstream systems such as Grafana, Zabbix, Splunk, ITSI, ServiceNow, event brokers, and automation engines can act on alerts without missing key context.

Outcome

For each CI type, the monitoring payload should make it possible to answer five questions immediately:

1. What failed?
2. Where did it fail?
3. How severe is it?
4. What data is required to diagnose or heal it?
5. Is the item safe to auto-remediate?

Core Design Principles

1. Every alert must identify the target unambiguously

Each alert should include a stable CI identifier and the exact affected component instance, not only a human-readable message.

2. Observability data must be operational, not only descriptive

Fields must support action. A message such as "disk full" is not enough if the payload does not tell us which filesystem, which host, current usage, and the threshold crossed.

3. Auto-heal must be gated by explicit safety metadata

Automation should not infer whether an action is allowed. The alert payload or enrichment layer should declare whether auto-heal is permitted and what script or runbook can be used.

4. Manual heal and auto-heal require different levels of precision

Manual response can tolerate narrative context. Auto-heal requires structured fields with deterministic names and values.

Standard Cross-CI Field Contract

The following fields should exist for all CI types, either in the original monitoring event or via enrichment before the alert reaches automation.

Field	Why it matters
ciType	Identifies the CI category such as service, disk, VM, interface, database, or certificate.
ciId	Stable unique identifier used by CMDB, ITSM, or automation.
ciName	Human-readable CI name.
host	Hostname or node where the issue occurred.
environment	Dev, QA, prod, pilot, or other operational scope.
region	Geographic or logical region for routing and compliance.
platform	Windows, Linux, network appliance, VMware, Azure, AWS, Kubernetes, and so on.
serviceOwner	Owning team or resolver group.
severity	Priority or impact classification.

Field	Why it matters
eventState	Problem, update, recovery, resolved, suppressed, maintenance.
firstSeen	First detection timestamp.
lastSeen	Last update timestamp.
alertRuleId	Monitoring rule or trigger identity.
alertSource	Zabbix, Grafana, Splunk, Azure Monitor, SNMP, and so on.
metricName	Name of the metric or signal that breached.
metricValue	Current observed value.
threshold	Trigger threshold or comparator.
correlationId	Event grouping key across tools.
runbookId	Linked manual recovery guide.
autoHealEligible	Explicit true or false flag for automation.
autoHealPolicy	Which automation policy or script family is allowed.
maintenanceFlag	Prevents action during approved maintenance windows.
changeContext	Current change ticket, rollout, or deployment context if applicable.

CI Requirements Matrix

The table below lists 20 common CI types and the most important fields needed for a fully working observability and healing model.

#	CI Type	Key observability fields	Manual heal data required	Auto-heal script inputs
1	Service	serviceName , serviceStatus , startupType , host , processId , port , dependencyServices , lastRestartTime	recent service logs, dependent service state, recent config changes, host health,	host , serviceName , expected state, allowed restart method, retry count, timeout, dependency order

#	CI Type	Key observability fields	Manual heal data required	Auto-heal script inputs
			maintenance status	
2	Disk / Filesystem	filesystem , mountPoint , driveLetter , usedPercent , freeBytes , inodePercent , deviceName , host	top directories, growth trend, recent cleanup jobs, file handle pressure, application owner	host , filesystem , cleanup profile, threshold breached, minimum free target, exclusion paths, safe-delete mode
3	Virtual Machine	vmName , powerState , guestState , cpuPercent , memoryPercent , datastore , hypervisor , host	console reachability, recent reboot history, patch state, VM tools status, attached disks, backup state	vmName , virtualization platform, action type, guest credentials reference, stop-start permission, health wait timeout
4	Network Interface	interfaceName , adminState , operState , speed , duplex , errorRate , discardRate , deviceName	link peers, VLAN, recent flaps, utilization trend, change window, config drift	deviceName , interfaceName , desired admin state, bounce permission, cooldown timer, escalation threshold
5	CPU Resource	host , cpuPercent , loadAverage , runQueue , stealTime , topProcesses , coreCount	process offenders, recent workload change, batch window, contention source, thermal or hypervisor context	host , remediation profile, process kill allowlist, scale trigger, CPU threshold, observation window
6	Memory Resource	host , memoryUsedPercent , availableMemory , swapUsage , pagingRate ,	memory leak indicators, process growth trend, recent deploys, swap	host , target process or service, safe restart order, free-memory threshold, memory cleanup mode

#	CI Type	Key observability fields	Manual heal data required	Auto-heal script inputs
		topConsumers , oomEvents	pressure, restart candidates	
7	Process	processName , processId , host , state , cpuPercent , memoryPercent , startTime , commandLine	parent process, log path, last exit code, ownership, dependency mapping, crash frequency	host , processName , start command, stop command, credential reference, expected instance count
8	Application Endpoint	applicationName , url , httpStatus , latencyMs , errorRate , availability , backendPool , environment	synthetic check history, backend dependency status, recent releases, certificate validity, DNS health	applicationName , url , health probe, restart target, rollback option, canary status, timeout
9	Database Instance	dbName , dbType , instanceName , listenerStatus , connections , replicationLag , storageUsed , host	blocking queries, failover status, backup freshness, transaction log growth, maintenance job state	dbName , instance identifier, service restart permission, failover target, credential secret ref, drain mode
10	Database Listener / Port	listenerName , host , port , status , acceptRate , connectionFailures , tlsState	port check history, firewall changes, process binding, certificate and cipher status	host , listenerName , port , restart action, validation probe, rollback condition
11	Kubernetes Pod	cluster , namespace , podName , podPhase , containerRestarts , node , image , readinessState	pod events, previous logs, deployment revision, HPA state, node	cluster , namespace , podName or label selector, restart policy, rollout object, grace period

#	CI Type	Key observability fields	Manual heal data required	Auto-heal script inputs
			pressure, recent rollout	
12	Kubernetes Node	cluster , nodeName , readyState , cpuPercent , memoryPercent , diskPressure , networkUnavailable	workload placement, taints, kubelet state, node events, draining status, infrastructure health	cluster , nodeName , cordon permission, drain policy, reboot method, rejoin validation
13	Container	containerName , containerId , image , exitCode , restartCount , cpuPercent , memoryPercent , host	container logs, orchestrator state, image version, secret mounts, volume health	containerId or workload name, restart command, image rollback tag, health probe, timeout
14	Load Balancer / VIP	vip , poolName , memberStatus , healthProbeStatus , failedMembers , latency , deviceName	affected pool members, probe configuration, change history, traffic drain status	deviceName , poolName , member identifiers, enable or disable action, probe path, validation check
15	Certificate / TLS Endpoint	certificateName , subject , issuer , expiryDate , daysToExpire , bindingTarget , thumbprint	bound services, renewal chain, ownership, key store path, change restrictions	certificateName , target endpoint, renewal source, import method, restart target list, validation URL
16	Scheduled Job / Batch	jobName , jobStatus , lastRunTime , lastSuccessTime , duration , exitCode , scheduler , host	job logs, upstream dependency status, business calendar, input file status, retry history	jobName , scheduler target, rerun mode, business-date parameter, dependency override flag, timeout

#	CI Type	Key observability fields	Manual heal data required	Auto-heal script inputs
17	Message Queue / Topic	queueName , depth , oldestMessageAge , consumerLag , deadLetterCount , broker , host	producer and consumer state, poison message sample, throughput trend, maintenance window	queueName , broker endpoint, purge or replay mode, consumer restart target, max drain batch
18	Storage Share / NAS / Object Store	shareName , bucketName , path , latency , availability , capacityUsed , iops , endpoint	mount clients, permission issues, recent quota changes, backend health, network path	storageTarget , path or share, reconnect method, remount option, cleanup threshold, validation probe
19	DNS Record / Resolver	recordName , recordType , resolvedValue , ttl , resolverStatus , lookupLatency , zoneName	authoritative server state, propagation status, recent change, dependency map	zoneName , recordName , desired value, rollback value, DNS provider target, propagation wait
20	Firewall / Security Rule / ACL	deviceName , ruleName , policyName , hitCount , adminState , lastChange , sourceZone , destinationZone	blocked flow details, related ticket, recent policy push, affected services, compliance owner	deviceName , policyName , ruleName , temporary override flag, expiration time, approval reference

Recommended Data Model Per Alert

Every actionable alert should carry four layers of information.

A. Identification

- CI type

- CI ID
- CI name
- exact affected subcomponent
- host, node, cluster, or device name

B. Detection Context

- metric name and current value
- threshold and comparator
- alert rule or trigger expression
- severity and impact
- first seen, last seen, and event count

C. Operational Context

- owning team
- business service or application
- environment and region
- maintenance state
- change or deployment context
- dependencies and parent service

D. Healing Context

- manual runbook ID
- auto-heal eligibility flag
- approved automation policy
- required script parameters
- rollback or validation criteria
- escalation path if automation fails

Minimum Auto-Heal Safety Controls

Auto-heal should only execute when the following metadata exists and is valid.

Control	Requirement
Target validation	The script can identify exactly one intended target.

Control	Requirement
Action approval	The CI type and resolver policy explicitly allow the action.
Maintenance awareness	The item is not in an active maintenance window unless approved.
Dependency awareness	Required upstream or downstream dependencies are known.
Retry policy	Maximum attempts and cooldown are defined.
Validation check	Post-action health verification is available.
Rollback rule	Reversal or escalation path exists if healing fails.
Auditability	Correlation ID, actor, action, and result are logged.

Suggested Payload Structure

```
{
  "ciType": "service",
  "ciId": "srv-win-ops-001:Spooler",
  "ciName": "Windows Print Spooler",
  "host": "server01.example.net",
  "environment": "prod",
  "severity": "high",
  "eventState": "problem",
  "alertSource": "grafana",
  "alertRuleId": "win-service-down",
  "metricName": "service_status",
  "metricValue": "down",
  "threshold": "expected=running",
  "observability": {
    "serviceName": "Spooler",
    "startupType": "Automatic",
    "dependencyServices": ["RPCSS"],
    "lastRestartTime": "2026-03-13T08:24:00Z"
  },
  "manualHeal": {
    "runbookId": "RB-WIN-SVC-001",
    "logLocation": "C:/Windows/System32/winevt/Logs/System.evtx",
    "ownerGroup": "IT_INFRA_CLS_UNIFIED-L1_OPS"
  },
  "autoHeal": {
    "autoHealEligible": true,
    "autoHealPolicy": "restart-windows-service",
    "inputs": {
      "host": "server01.example.net",
      "serviceName": "Spooler",
      "retryCount": 2,
      "timeoutSeconds": 90
    },
    "validation": {
      "expectedState": "running",
      "checkAfterSeconds": 30
    }
  },
  "correlationId": "evt-20260313-000145"
}
```

Implementation Guidance

Monitoring tools should provide

- raw metric and alert-rule details
- target-specific labels such as service name, filesystem name, interface name, pod name, or queue name
- timestamps and event lifecycle state

Enrichment layers should provide

- CI identifiers from CMDB
- owner and assignment data
- environment, platform, and business service mapping
- runbook and auto-heal policy references

Automation platforms should require

- structured inputs only
- an explicit `autoHealEligible` gate
- post-remediation validation result
- full audit logging

Priority Recommendation

If the implementation must be phased, start with the CI types that usually provide the fastest operational value and safest remediation profile:

1. Service
2. Disk / Filesystem
3. Process
4. Virtual Machine
5. Application Endpoint
6. Scheduled Job
7. Kubernetes Pod
8. Certificate / TLS Endpoint

Final Recommendation

The monitoring payload contract should be treated as an integration standard, not as a dashboard convenience. If a field is required by a runbook or a healing script, it must be present as a structured attribute in the event stream and not hidden only inside free-text descriptions.

That is the difference between alerts that are visible and alerts that are actionable.